Owen Ritchie

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Education

Honours Bachelor of Science, Computer Science — *University of Ottawa* **Sep 2022–Apr 2027**

- Dean's List. Relevant coursework: Modelling & Simulation, Design & Analysis of Algorithms, Artificial Intelligence & Cybersecurity
- Honours project: CSI 4900 student-professor project portal built with TypeScript, React, C#, Terraform & Supabase
- Will be pursuing a part-time Master of Data Science at a different institution following graduation

Work Experience

Data & Analytics Engineer Intern — *Kinaxis* **May 2026–Aug 2026**

- Built pipelines for **3** Google Analytics properties, including kinaxis.com, handling **250k+** events daily, using dbt, GCS & BigQuery
- Migrated **~90M** rows of Eloqua campaign data between Snowflake & BigQuery, ensuring zero downtime, using dbt & SQL
- Optimized **11** dataflows, cutting **100k+** CU-hours of yearly Fabric compute (**~$18k in savings**), using BigQuery, dbt & Power BI
- Developed a Win/Loss classifier to assign labels to Salesforce opportunities based on SDR comments, using BigQuery ML & Gemini
- Audited 110+ Grafana reports, flagging 80%+ as redundant and adding better filtering to the most-used ones, using Python & SQL

Data & Analytics Engineer Intern — *Kinaxis* **Sep 2025–Apr 2026**

- Engineered a customer footprint tool aggregating **500M+** parts, scenarios & users for the incoming CEO, using BigQuery & Datadog
- Built a Chrome extension to optimize pipeline cost against committed spend at our hackathon, using Vertex AI, TypeScript & GCP
- Migrated **10+** years of ServiceNow & LogicMonitor data from Postgres to GCS, eliminating **2+** days of manual reporting each month
- Built a report aggregating Jira tasks & epics to estimate feature velocity across **300+** engineers in Product for the CPO, using SQL
- Redesigned Kinaxis' Cloud FinOps report to handle project-level commit tracking for the VP of Cloud Services, using Power BI

Data Analytics Intern — *Global Affairs Canada* **Jan 2025–Jul 2025**

- Curated and validated **5+** years of cross-institutional data backing **50+** statistics in Canada's G7 Accountability Report, using Python
- Reduced API response times by **99.8%** by implementing multithreading to ingest data from the World Bank, OECD, UN and WEF
- Built a development indicators tool for **10+** Directors to model the impact of policy across international aid funding scenarios
- Delivered reports on the Sahel and CARICOM covering **25+** metrics, including HDI and electrification, using Scikit-learn & Matplotlib

Data Science Intern — *Poisson Consulting Ltd.* **May 2023–Sep 2023**

- Analyzed **500k+** records using R and Tidyverse to support fish population revitalization efforts in the Lower Columbia River
- Mapped **3k+** salmon redds from drone surveys in QGIS, allowing clients to optimize dam levels for increased egg survival
- Volunteered as a TA at Columbia Mountains Institute of Applied Ecology, teaching **15** students to apply R in ecology & biology

Personal Projects

owenritchie/data — *dbt, SQL, Python, GitHub Actions, Neon (Postgres)* **2025–Present**

- Built an analytics repository that automates ingestion (Python) and transformation (dbt) for my personal projects
- Powers Spotify Analytics (React, my listening history) and OC Transpo Analytics (Streamlit, Ottawa bus fleet monitoring)

Mill — *Next.js, React, TypeScript, Tauri, Rust, Drizzle, Claude SDK* **2026–Present**

- Built a full-stack macOS project planner with a node-based canvas and chat-based planning via Ollama & Claude models
- Packaged the app as a .dmg with Tauri & a GitHub Actions release workflow, storing data locally in libSQL via Drizzle

CSI 4900 Project Portal — *TypeScript, SQL, Supabase, AWS S3, Terraform* **2026–Present**

- Designed the Supabase Postgres schema for a student-professor project portal used by **200+** students & supervisors each semester
- Implemented row-level security and a custom JWT hook enforcing user roles across projects, inquiries and attachments
- Wrote **320+** pgTAP tests covering schema, RLS policies and the authentication hook

Technical Skills

Languages: Python, SQL, TypeScript, R, Java, JavaScript, HTML & CSS, Bash/PowerShell

Frameworks & Libraries: dbt, Pandas, Scikit-learn, Apache Spark, TensorFlow, Node.js, OpenAI, Vertex AI, Claude SDK, React, Astro.js

Infrastructure: BigQuery, Snowflake, GCS, AWS S3, Cloud Run, GitHub Actions, Postgres, Supabase, Airflow, Linux, Git, Docker

Reporting: Power BI, Looker Studio, Grafana, Datadog, Metabase, R Markdown, Jupyter

Active Government of Canada Secret clearance (Level II).